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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (VDE) (CBCS) Degree Examinations

December 2024- January 2025

22VDE121 – ADVANCED DIGITAL SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

1. a) Design a one-input one-output sequence detector that produces an output value 1 every time the sequence 0101 is detected and an output value 0 at all other times. For example, if the input sequence is 010101 then the corresponding output sequence is 000101.
- b) Draw the Compatibility graph for **Machine M1** shown in **Fig. 1(b)**. Also find the closed set of compatibles using this compatibility graph and hence find a minimal machine that covers M1.

Marks BT* CO* PO*

10 L*3 1 3

PS	NS, z			
	I ₁	I ₂	I ₃	I ₄
A	—	—	E, 1	—
B	C, 0	A, 1	B, 0	—
C	C, 0	D, 1	—	A, 0
D	—	E, 1	B, —	—
E	B, 0	—	C, —	B, 0

Fig. 1(b) Machine M1

10 L3 2 3

2. a)

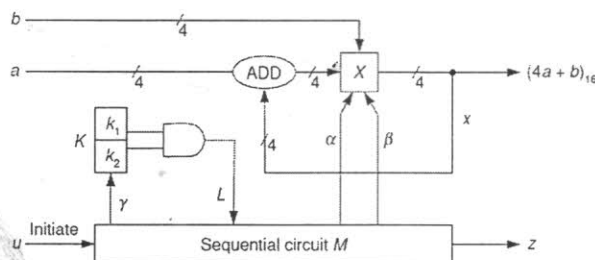


Fig. 2(a): A system to compute (4a + b) modulo 16

For the system shown in **Fig. 2(a)**, draw a state diagram, write the transition table, find the expressions for output and next states along with the logic diagram.

10 L3 1 3

- b) When are two states of a machine said to be k-distinguishable and k-equivalent. Refer to **Machine M2** shown in **Fig. 2(b)** and explain how to find the distinguishing sequence for the pair (A, B) and (A, E). Also give an example for equivalent states.

PS	NS, z	
	x = 0	x = 1
A	E, 0	D, 1
B	F, 0	D, 0
C	E, 0	B, 1
D	F, 0	B, 0
E	C, 0	F, 1
F	B, 0	C, 0

Fig. 2(b): Machine M2

10 L3 2 3

3. a) Draw the State diagram for a modulo-8 binary counter. Implement modulo-8 binary counter using T flip flop.
- b) Define Merger graph of an n-state Machine M. Machine M3 is shown in Fig. 3(b). Draw the Merger graph for M3. Also mention all the compatible pairs and arrive at maximal compatibles for machine M3.

PS	NS, z			
	I_1	I_2	I_3	I_4
A	—	C, 1	E, 1	B, 1
B	E, 0	—	—	—
C	F, 0	F, 1	—	—
D	—	—	B, 1	—
E	—	F, 0	A, 0	D, 1
F	C, 0	—	B, 0	C, 1

Fig. 3(b) Machine M3

4. a) Draw the map and gate network for the function $T(x,y,z) = \sum(2,3,5,7)$. Explain single input change static hazard. Also explain how to make it hazard free and draw the hazard free network.
- b) Consider the primitive flow table shown in Table 4b. Explain the method to reduce the flow table and hence find the reduced flow table. Write the equivalent flow table with state assignment. Also find the implication table corresponding to this flow table.

Table 4b: Primitive flow table

	x_1x_2			
	00	01	11	10
S_0	$(S_0), 0$	—	—	S_1
S_1	S_2	—	—	$(S_1), 1$
S_2	$(S_2), 1$	S_5	—	S_3
S_3	S_4	—	—	$(S_3), 1$
S_4	$(S_4), 1$	S_6	—	—
S_5	S_0	$(S_5), 1$	—	—
S_6	S_2	$(S_6), 1$	—	—

5. a) Consider the flow table given in Table 5a. Write the excitation table and output table. Construct the asynchronous circuit using logic gates by finding the excitation and output functions using K map.

Table 5a: Flow table

	x_1x_2			
	00	01	11	10
A	(A),0	(A),0	(A),0	C
B	(B),1	D	(B),1	C
C	A	(C),1	B	(C),1
D	—	A	—	—

- b) What are static and dynamic hazards? With the help of an example circuit, explain hazards in asynchronous sequential circuits and combinational circuits.

10 L3 3 3

10 L3 3 3

6. a) What is a race condition in a fundamental model circuit? Comment on critical and non-critical race condition in a fundamental model circuit. Considering the flow table shown in Table 6a, write the excitation table and also illustrate the critical race condition for this flow table.

Table 6a: Flow table

	x_1x_2			
	00	01	11	10
A	(A)	C	(A)	C
B	A	(B)	(B)	D
C	D	(C)	B	(C)
D	(D)	B	A	(D)

- b) Explain the dynamic 1→0 hazard for the circuit with Boolean expression $f = x'y + wx + yz' + wz + wy$. Redesign it to make it a Logic-hazard-free network.

10 L3 3 3

10 L3 3 3

Unit – III

7. a) Explain the terms bound line, free line, head line and technique multiple backtracks used in FAN algorithm. Illustrate the application of FAN algorithm by deriving a test for the fault Z s-a-0 in Fig. 7(a) considering any sequential circuit as an example.

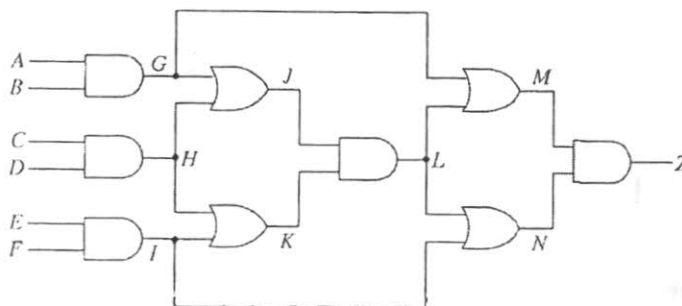


Fig. 7(a): Circuit under test

10 L3 4 3

- b) Illustrate the test generation process by deriving a test sequence for the fault α s-a-0 in the sequential circuit shown in Fig. 7(b).

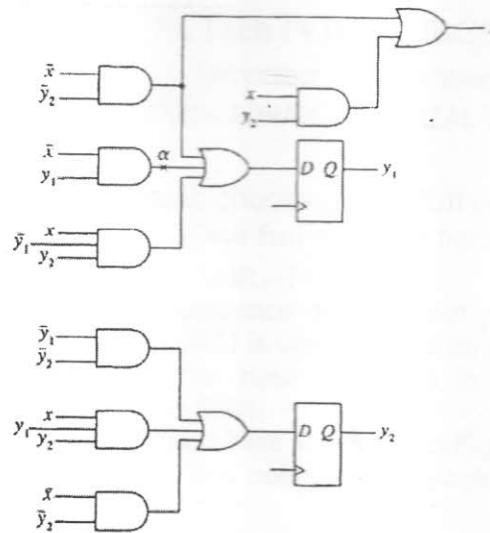


Fig. 7(b): A sequential circuit

10 L3 5 3

8. a) Write the different steps involved in PODEM algorithm. Illustrate the application of PODEM by deriving a test for fault l s-a-1 in the circuit shown in Fig. 8(a).

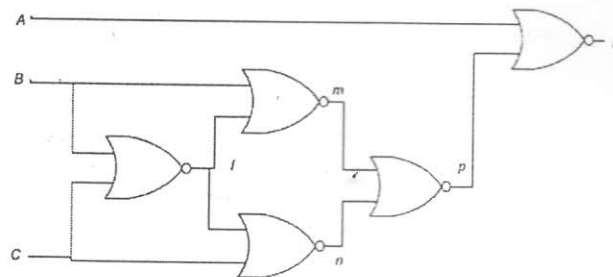


Fig. 8(a): Circuit under test

10 L3 5 3

- b) With the help of an example for each, explain 3 types of cubes used in D algorithm.

10 L3 4 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
